Kisan Mitra AI - Smart Crop Recommendation Report

Farmer Email:	DIVYA.MUNNURU@GMAIL.COM	**Season:**	Kharif
State:	Punjab	**District:**	Ludhiana
Soil Type:	Black Soil	**Irrigation:**	Medium
Water Source:	Medium	**Farm Size:**	2 Acres
Previous Crop:	rice	**Date:**	16 June 2026

Seasonal Suitability Analysis

Overall Suitability Score: 88%

Agricultural analysis for Ludhiana, Punjab during Kharif (Monsoon/June-October) season. The region's Black Soil is excellent for cultivation. With Medium irrigation availability and Medium water source, the farm is well-positioned for moderate crops.

General Advice: For Black Soil in Ludhiana, ensure proper soil testing before sowing. Maintain balanced irrigation schedules. Consider crop rotation with legumes to replenish soil nitrogen.

■■ Critical Alerts

• Monitor local weather advisories for Punjab during Kharif season for unexpected rainfall or temperature changes.

• Ensure seed treatment with fungicide before sowing to protect against soil-borne diseases common in Black Soil.

• Previous crop was rice. Practice crop rotation to break pest and disease cycles.

Recommended Crops

Maize (Corn) (Suitability Score: 90% | Water: Medium | Demand: High | Growing Period: 90-110 days)

Why Recommended:	Fast-growing cereal crop with diverse uses (food, feed, industrial starch). Adapts well to various soil types and gives good returns in Kharif season. Well-suited for Ludhiana, Punjab region with Black Soil.
Water Management:	Maize needs moderate water with critical irrigation at tasseling and grain-filling stages. Avoid waterlogging as it damages root systems. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply DAP (50 kg/acre) and MOP (30 kg/acre) as basal. Top dress with Urea (40 kg/acre) at knee-high stage and tasseling.
Expected Yield:	**20-30 quintals/acre**

Cotton (Suitability Score: 88% | Water: Medium | Demand: High | Growing Period: 150-180 days)

Why Recommended:	Major commercial cash crop with strong market linkages. Performs best in black cotton soils with warm temperatures above 20°C. Well-suited for Ludhiana, Punjab region with Black Soil.
Water Management:	Cotton requires moderate and well-timed irrigation. Critical water needs are during flowering and boll formation. Excess water causes boll rot. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply FYM (5 tonnes/acre), DAP (50 kg/acre) as basal, Urea (30 kg/acre) at squaring stage, and spray micronutrients (Boron, Zinc).
Expected Yield:	**8-12 quintals/acre (seed cotton)**

Pigeon Pea (Arhar/Tur Dal) (Suitability Score: 85% | Water: Medium | Demand: High | Growing Period: 140-180 days)

Why Recommended:	Essential pulse crop for protein security. Fixes atmospheric nitrogen, improving soil fertility for subsequent crops. High MSP support from government. Well-suited for Ludhiana, Punjab region with Black Soil.
Water Management:	Pigeon pea is drought-tolerant but responds well to 2-3 irrigations during flowering and pod development if rainfall is insufficient. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply Rhizobium seed treatment, DAP (40 kg/acre) as basal. Minimal nitrogen needed due to biological fixation. Spray 2% Urea foliar at flowering.
Expected Yield:	**6-10 quintals/acre**

Soybean (Suitability Score: 85% | Water: Low | Demand: High | Growing Period: 90-110 days)

Why Recommended:	Called 'Golden Bean' for its high protein and oil content. Excellent for rain-fed farming with short duration. Fixes nitrogen and improves soil health. Well-suited for Ludhiana, Punjab region with Black Soil.
Water Management:	Soybean is well-suited for rain-fed conditions and can manage with minimal supplemental irrigation. Ensure good drainage as it is sensitive to waterlogging. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Rhizobium + PSB seed inoculation. Apply DAP (40 kg/acre) as basal. Avoid excessive nitrogen as it inhibits nodulation.
Expected Yield:	**8-12 quintals/acre**

■ High Risk / Unsuitable Crops

Wheat: Wheat is a cool-season Rabi crop requiring temperatures of 15-25°C. Kharif season's high heat (30-40°C) and humidity cause poor germination, excessive vegetative growth, and severe rust/blight diseases.

Mustard: Mustard requires cool, dry winters for proper seed formation. Planting in Kharif causes excessive vegetative growth, flower drop due to heat, and white rust/Alternaria blight in humid conditions.